

UHRP High-Risk Pregnancy Tracking System

Technical Architecture, Process Flow & Schema Documentation

Prepared for the Clinical and Operational Teams • August 2026

1. System Overview & Context

The Unified High-Risk Pregnancy (UHRP) Tracking System is an integrated data and analysis pipeline designed to monitor pregnant women identified with high-risk conditions (such as Severe Anemia, PIH, GDM, and Sickle Cell Disease) across multiple blocks and facility centers. The system aggregates survey responses from three distinct Google Forms, executes data normalization and fuzzy patient linkage, computes gestational-age-based clinical alerts, and compiles a comprehensive, interactive client dashboard.

The primary objective of the system is to prevent maternal and neonatal complications by giving coordinators, doctors, and program managers visibility into each patient's clinical trajectory from identification to follow-up, and finally to delivery outcome.

2. Integrated Data Pipeline Flow

The pipeline runs locally through a Python compilation engine and outputs a fully static, standalone HTML dashboard that requires no server infrastructure to distribute or open. The process flow is detailed below:

Step 1: Data Export: Excel response sheets are downloaded from Google Forms (Form A - Identification, Form B - Follow-up, and Form C - Delivery Outcomes) and placed in the project working folder.

Step 2: Schema Alignment & Column Rename: The Python script `generate_analysis_data.py` loads the files via Pandas, mapping any updated column headers or regional translations to standard internal keys.

Step 3: Patient Normalization: MP ID values are stripped of decimal format markers and whitespaces. Mobile numbers are normalized to extract digit sequences.

Step 4: Relational Database Construction: Identification cases are loaded as core patient profiles. Follow-up visits (Form B) and Delivery records (Form C) are then matched to their respective mother using standard ID checks and fuzzy name/husband-name metrics.

Step 5: Alert Calculations: Gestational age calculations are triggered relative to the execution date to compile follow-up targets. Delivery trajectories are evaluated for resolution or worsening.

Step 6: Dashboard Compilation: A compiled JSON dataset payload is injected directly into `index.html` (template) to output a single shareable HTML file `uhrp_dashboard_shareable.html`.

3. Data Schemas & Structure

Below are the essential columns and data formats compiled from the three Google Form sources:

Form A: Identification Record (Enrollment)

Field/Column Header	Format / Type	Description / Key Values
MP ID	String (11 digits)	Unique Mother ID. Key linkage parameter.
ANC Name / Husband Name	String	Demographics for identification & fuzzy matching.
LMP / EDD	Date (DD-MM-YYYY)	Used for gestational age & follow-up calculations.
HRP Categories Identified	Comma-separated list	Severe Anemia, PIH, GDM, SCD, Teenage, etc.
Hb / BP / Sugar at ID	Numeric / String	Baseline clinical values at enrollment.

Form B: Follow-up Visit Record

Field/Column Header	Format / Type	Description / Key Values
MP ID / Visit Number	String / Integer	Links visit to enrollment; tracks visit count.
Current Hb / BP / Sugar	Numeric / String	Follow-up clinical metrics to evaluate progression.
Medication Adherence	String	Tracks whether patient is taking IFA/calcium/antihypertensives.
Overall HRP Status	String	Controlled, Worsened, Stable, or Improved.

Form C: Delivery Outcome Record

Field/Column Header	Format / Type	Description / Key Values
Date & Place of delivery	Date / String	Tracks timing and venue (Home, PHC, CHC, DH).
Type / Mode of delivery	String	Normal vaginal delivery, Elective/Emergency LSCS.
Baby Born Alive / Birth Weight	String / Numeric	Critical newborn outcome parameters.
Baby Transferred / Reason	String / String	Tracks SNCU/NICU admission indicators.

4. Key Analysis Algorithms

4.1. Relational Matching & Fuzzy Linkage

To connect isolated entries across Google Forms and build cohesive patient history timelines, the compilation script employs a multi-tiered linkage algorithm:

- 1. Exact MP ID Matching:** Attempts to bind Form B and Form C records to Form A using normalized MP IDs (whitespaces and decimal strings removed).
- 2. Fuzzy Name & Husband Matching:** If the MP ID is blank or typed in the follow-up/delivery sheet, the script matches using standard lowercase name strings and husband names.
- 3. Manual Linkage Overrides:** For highly anomalous transcription errors (e.g. dummy MP IDs like '00000000000' or missing names), hardcoded relational rules are defined to preserve data integrity.
- 4. Unmatched Outcome Isolation:** Any Form C record that cannot be linked using these three tiers is categorized as an 'Unmatched Outcome' and placed in a dedicated, isolated tab on the dashboard, highlighting data entry gaps.

4.2. Gestational Age & Follow-up Alert Calculations

The follow-up due list is dynamically recalculated relative to the compilation date. The algorithm evaluates:

- 1. Active Status:** Filters out mothers who have already delivered (linked Form C record is present).
- 2. Gestational Age Calculation:** Gestational Age (GA) is computed relative to the script execution date: $GA = (\text{Execution Date} - \text{LMP Date}) / 7$.
- 3. Follow-up Target:** A patient is flagged as 'Due for Follow-up' if their current GA falls between **32 to 34 weeks** and no Form B visit has been recorded in the database. Once generated, this list updates automatically.

4.3. Delivery Trajectory Progression Metrics

To evaluate quality of care and condition resolution, the script analyzes changes in overall status between identification and delivery. Progression is categorized as follows:

Improved: Clinical indicators or HRP status transitioned from active risk to normal/negative state (e.g., Hb rose above 11 g/dL, or PIH was resolved).

Controlled / Stable: Risk factor was kept stable throughout pregnancy and delivery (e.g. blood pressure stayed under 140/90 mmHg for a PIH case).

Worsened: Clinical indicators deteriorated (e.g., Hb fell, PIH progressed to severe, or GDM became uncontrolled).

5. Efficiency Opportunities & Recommended Flows

To further streamline the data pipeline, improve tracking accuracy, and reduce manual intervention, we suggest the following architectural updates:

Current Flow	Proposed Improvement	Impact & Efficiency Gain
Manual download of Excel files from Google Drive.	Integrate the Google Sheets API directly into <code>generate_analysis_data.py</code> .	Automation: Eliminates manual exports. The compiler fetches fresh responses in real-time with a single click/command.
Manual typo corrections & hardcoded matching overrides in Python script.	Implement Form Validation rules in Google Forms (e.g., regex checks forcing exactly 11 digits for MP ID).	Data Quality: Prevents transcription errors (like missing digits or dummy zeros) at the source of entry.
Local execution of compiler script by team members.	Set up a scheduled task (local Windows Task Scheduler) or run as a free GitHub Action with GitHub Pages.	Zero-touch: The dashboard re-compiles and updates automatically every evening without requiring Python local setups.
Separate spreadsheets for each pregnancy form.	Create a relational database (SQLite/PostgreSQL) or use AppSheet as the data collection interface.	Robustness: Enforces relational integrity. A follow-up visit or delivery cannot be submitted without selecting a valid enrolled patient.

6. Conclusion & Next Steps

By dynamically isolating unmatched deliveries and implementing robust data cleaners, the UHRP tracking dashboard now displays exactly 12 matched delivery outcomes alongside an actionable list of 5 unmatched outcomes. Addressing the data capture quality in the field using form validation is the highest priority next step to eliminate manual data cleaning overrides.